

Prophet Model

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PROPHET MODEL FOR DENGUE FORECASTING

Load Libraries

```
library(prophet)
library(readr)
library(tidyverse)
library(Metrics)
```

LOAD DATA

```
dengue_data <- read_csv(
  "C:/Users/ADMIN/Desktop/Time Series Session/Datasets/Dengue.csv"
)
```

DATA EXPLORATION

View structure

```
glimpse(dengue_data)
```

```
## Rows: 2,557
## Columns: 2
## $ ds <date> 2019-01-01, 2019-01-02, 2019-01-03, 2019-01-04, 2019-01-05, 20
19-0...
## $ y <dbl> 16, 14, 20, 20, 10, 11, 16, 17, 20, 18, 23, 21, 18, 22, 22, 20,
25,...
```

Summary statistics

```
summary(dengue_data)
```

```
##           ds                y
## Min.      :2019-01-01   Min.   : 5.00
## 1st Qu.:2020-10-01   1st Qu.:37.00
## Median :2022-07-02   Median :51.00
## Mean      :2022-07-02   Mean    :50.45
## 3rd Qu.:2024-04-01   3rd Qu.:64.00
## Max.      :2025-12-31   Max.    :94.00
```

Missing values

```
colSums(is.na(dengue_data))
```

```
## ds y
## 0 0
```

DATA PREPARATION

Convert date column

```
dengue_data$ds <- as.Date(dengue_data$ds)
```

Ensure target variable is numeric

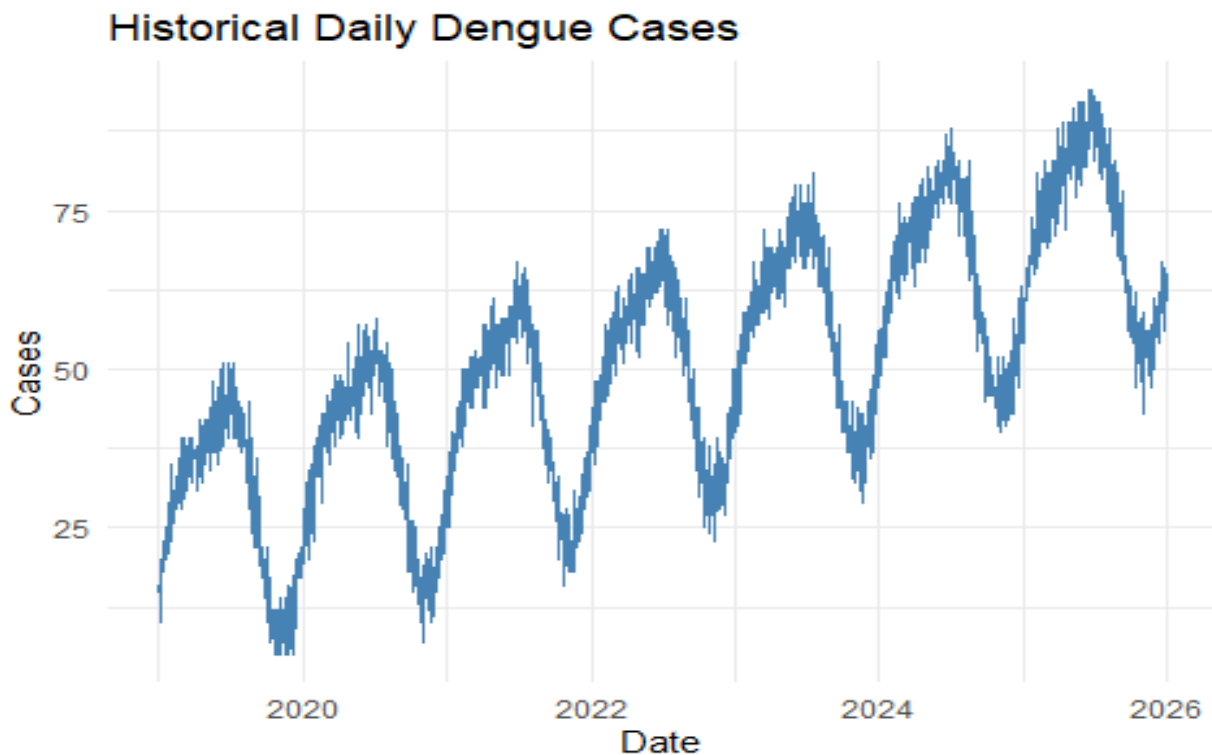
```
dengue_data$y <- as.numeric(dengue_data$y)
```

Sort by date

```
dengue_data <- dengue_data %>%  
  arrange(ds)
```

VISUALIZE HISTORICAL DATA

```
ggplot(dengue_data,  
       aes(x = ds,  
           y = y)) +  
  geom_line(color = "steelblue") +  
  labs(  
    title = "Historical Daily Dengue Cases",  
    x = "Date",  
    y = "Cases"  
  ) +  
  theme_minimal()
```



```

## Why I chose Prophet
## 1. Clear Seasonality
## Dengue cases often rise and fall during certain periods of the year.
## Long Historical Data. We have 2557 daily observations ≈ 7 years of data
## Prophet generally performs well when several years of history are available.
## 3. Trend Changes
## Disease cases can increase or decrease over time.
## Prophet automatically detects trend changes (changepoints) without requiring extensive manual modeling.
## 4. Easy Interpretation
## Prophet separates the forecast into:
## Trend
## Yearly seasonality
## Weekly seasonality
## making it easy to explain in reports and presentations.

## CHECK FOR MISSING DATES

all_dates <- seq(
  from = min(dengue_data$ds),
  to = max(dengue_data$ds),
  by = "day"
)

cat("Expected dates:", length(all_dates), "\n")
## Expected dates: 2557

cat("Actual rows:", nrow(dengue_data), "\n")
## Actual rows: 2557

## TRAIN-TEST SPLIT (80% TRAIN, 20% TEST)

split_index <- floor(0.80 * nrow(dengue_data))

train_data <- dengue_data[1:split_index, ]
test_data <- dengue_data[(split_index + 1):nrow(dengue_data), ]

cat("Training observations:", nrow(train_data), "\n")
## Training observations: 2045

cat("Testing observations:", nrow(test_data), "\n")
## Testing observations: 512

## The Prophet model was trained using 2,045 observations (80%) of the dataset.
## The remaining 512 observations (20%) were reserved for testing.

```

This allows the model's forecasting ability to be evaluated on unseen data.

FIT PROPHET MODEL

```
model <- prophet(  
  yearly.seasonality = TRUE,  
  weekly.seasonality = TRUE,  
  daily.seasonality = FALSE,  
  changepoint.prior.scale = 0.05  
)
```

```
model <- fit.prophet(model, train_data)
```

yearly.seasonality = TRUE → Captures yearly patterns in dengue cases.

weekly.seasonality = TRUE → Captures weekly variations in the data.

daily.seasonality = FALSE → Daily patterns are not modeled because the data does not require them.

changepoint.prior.scale = 0.05 → Controls how flexible the trend is; a small value produces a smoother trend.

fit.prophet(model, train_data) → Trains the Prophet model using the training dataset.

FORECAST TEST PERIOD

```
future_test <- make_future_dataframe(  
  model,  
  periods = nrow(test_data),  
  freq = "day"  
)
```

```
forecast_test <- predict(model, future_test)
```

make_future_dataframe() creates future dates for forecasting.

periods = nrow(test_data) tells Prophet to forecast the same number of days as the test dataset (512 days).

freq = "day" specifies that forecasts are generated daily.

future_test contains both historical and future dates.

predict(model, future_test) uses the trained Prophet model to generate forecasts for those dates and stores the results in forecast_test.

EXTRACT TEST PREDICTIONS

```
predictions <- forecast_test %>%  
  tail(nrow(test_data)) %>%  
  select(ds, yhat, yhat_lower, yhat_upper)
```

MODEL EVALUATION

```

RMSE <- rmse(
  actual = test_data$y,
  predicted = predictions$yhat
)

MAE <- mae(
  actual = test_data$y,
  predicted = predictions$yhat
)

MAPE <- mean(
  abs((test_data$y - predictions$yhat) /
    test_data$y),
  na.rm = TRUE
) * 100

R2 <- cor(test_data$y,
  predictions$yhat)^2
cat("\n")

cat("MODEL PERFORMANCE\n")

## MODEL PERFORMANCE

cat("-----\n")

## -----

cat("RMSE :", round(RMSE, 2), "\n")

## RMSE : 3.17

## On average, the model's predictions differ from the actual dengue cases by about 3.17 cases.
## A Lower RMSE indicates better predictive accuracy.
## Since dengue cases range from approximately 5 to 94 cases, an RMSE of 3.17 is very low.

cat("MAE :", round(MAE, 2), "\n")

## MAE : 2.51

## The model's forecasts differ from the actual values by about 2.51 cases on average.
## This indicates that prediction errors are small and the model is highly accurate.

cat("MAPE :", round(MAPE, 2), "%\n")

## MAPE : 3.83 %

```

The forecasts deviate from the actual dengue cases by only 3.83% on average.
A MAPE below 10% is generally considered highly accurate.
Therefore, the Prophet model demonstrates excellent forecasting performance.

```
cat("R-squared:", round(R2,4))
```

```
## R-squared: 0.9533
```

The model explains approximately 95.33% of the variation in dengue cases.
This indicates a very strong relationship between the predicted and actual values.
Values closer to 1 indicate better model fit.

ACTUAL VS PREDICTED

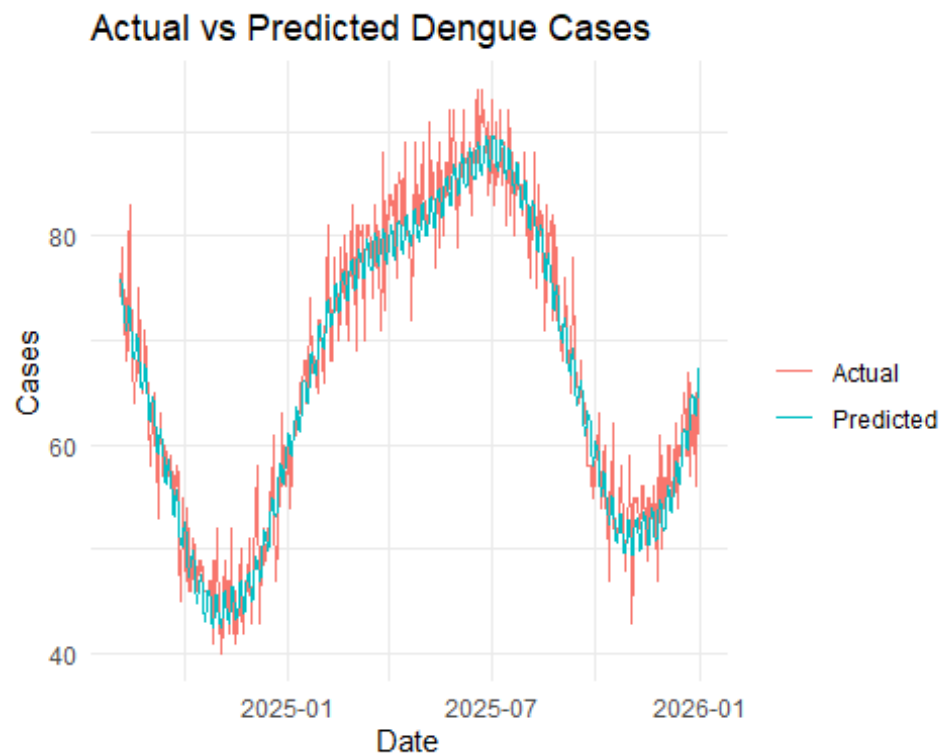
```
comparison <- data.frame(  
  Date = test_data$ds,  
  Actual = test_data$y,  
  Predicted = predictions$yhat  
)
```

```
ggplot(comparison,  
  aes(Date)) +
```

```
  geom_line(  
    aes(y = Actual,  
      color = "Actual")  
  ) +
```

```
  geom_line(  
    aes(y = Predicted,  
      color = "Predicted")  
  ) +
```

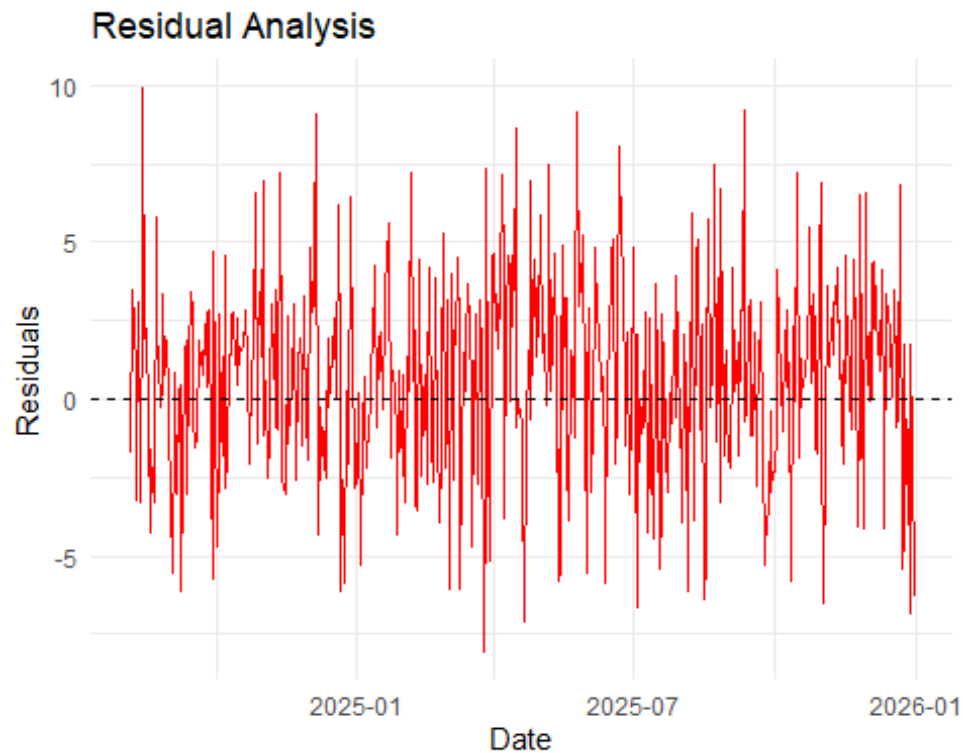
```
  labs(  
    title = "Actual vs Predicted Dengue Cases",  
    x = "Date",  
    y = "Cases",  
    color = ""  
  ) +  
  theme_minimal()
```



```
# RESIDUAL ANALYSIS

comparison$Residuals <-
  comparison$Actual -
  comparison$Predicted

ggplot(comparison,
  aes(Date, Residuals)) +
  geom_line(color = "red") +
  geom_hline(
    yintercept = 0,
    linetype = "dashed"
  ) +
  labs(
    title = "Residual Analysis",
    x = "Date",
    y = "Residuals"
  ) +
  theme_minimal()
```



```
## REFIT MODEL ON FULL DATASET
```

```
final_model <- prophet(  
  dengue_data,  
  yearly.seasonality = TRUE,  
  weekly.seasonality = TRUE,  
  daily.seasonality = FALSE,  
  changepoint.prior.scale = 0.05  
)
```

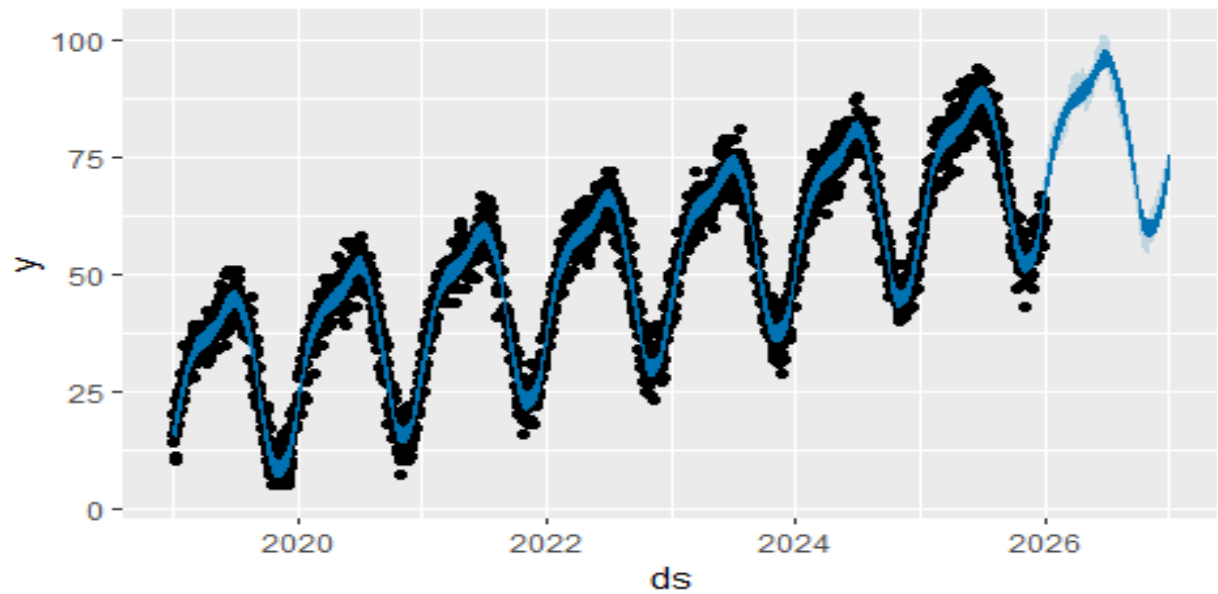
```
## FORECAST NEXT 365 DAYS
```

```
future <- make_future_dataframe(  
  final_model,  
  periods = 365,  
  freq = "day"  
)
```

```
forecast <- predict(  
  final_model,  
  future  
)
```

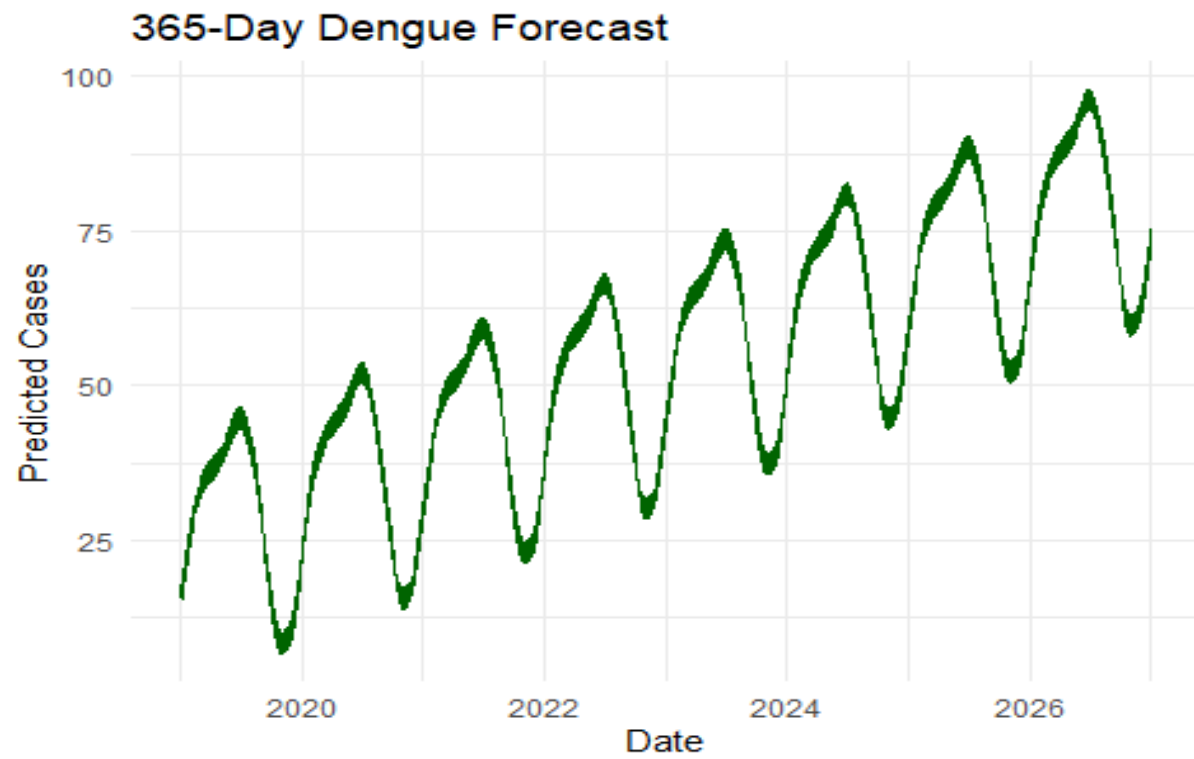
```
## FORECAST PLOT
```

```
plot(final_model, forecast)
```

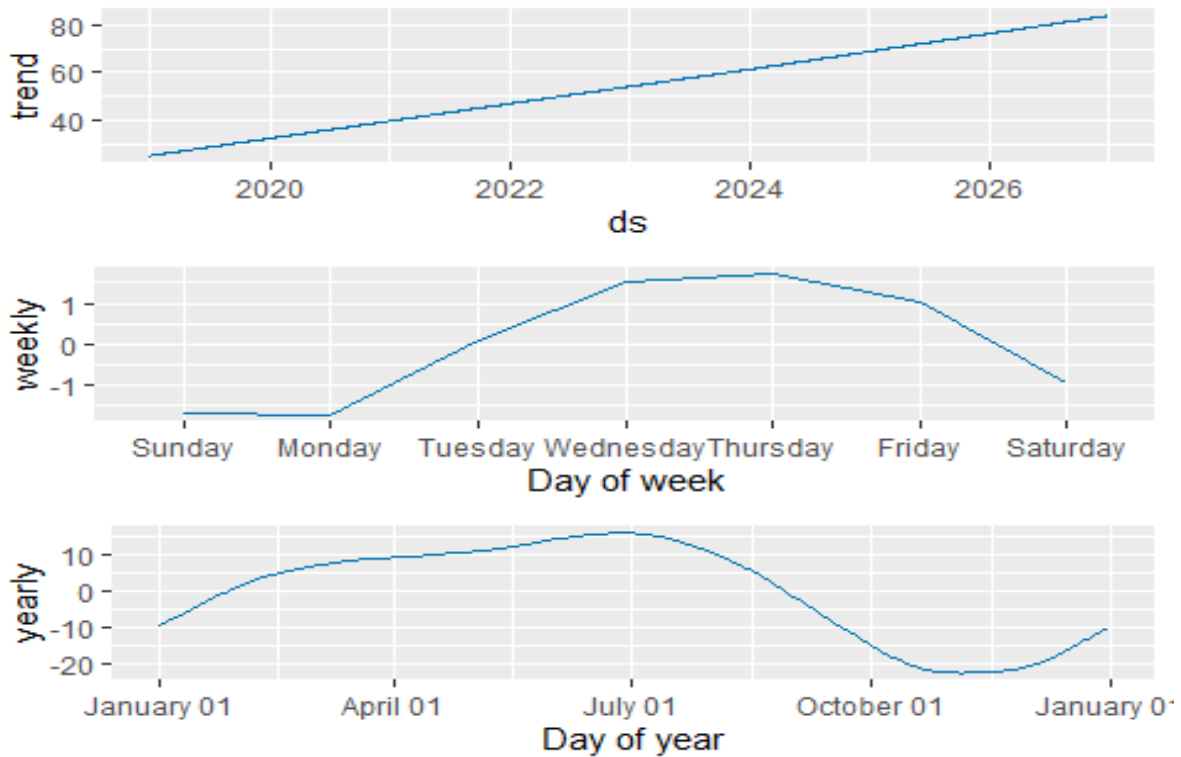
FORECASTED VALUES PLOT

```
ggplot(forecast,  
      aes(ds, yhat)) +  
  geom_line(color = "darkgreen") +  
  labs(  
    title = "365-Day Dengue Forecast",  
    x = "Date",  
    y = "Predicted Cases"  
  ) +  
  theme_minimal()
```



SEASONALITY COMPONENTS

```
prophet_plot_components(  
    final_model,  
    forecast  
)
```



FORECAST TABLE

```
forecast %>%
  select(
    ds,
    yhat,
    yhat_lower,
    yhat_upper
  ) %>%
  tail(20)
```

##	ds	yhat	yhat_lower	yhat_upper
## 2903	2026-12-12	64.65212	60.87046	68.30345
## 2904	2026-12-13	64.31429	60.82249	68.21876
## 2905	2026-12-14	64.68280	60.72978	68.50457
## 2906	2026-12-15	66.86799	62.45765	70.60005
## 2907	2026-12-16	68.72560	64.84665	72.59061
## 2908	2026-12-17	69.32326	65.70773	73.37244
## 2909	2026-12-18	69.05887	65.40178	72.93651
## 2910	2026-12-19	67.53321	63.79297	71.18542
## 2911	2026-12-20	67.24375	63.44634	71.03560
## 2912	2026-12-21	67.65235	63.80147	71.58125
## 2913	2026-12-22	69.86943	66.17409	73.68767
## 2914	2026-12-23	71.75090	67.81918	75.65573
## 2915	2026-12-24	72.36471	68.70905	76.32939
## 2916	2026-12-25	72.10919	68.46987	75.86302
## 2917	2026-12-26	70.58565	66.73934	74.41862

## 2918	2026-12-27	70.29215	66.46614	74.14853
## 2919	2026-12-28	70.69128	66.82751	74.45635
## 2920	2026-12-29	72.89418	69.15661	76.62696
## 2921	2026-12-30	74.75757	70.95069	78.55831
## 2922	2026-12-31	75.35021	71.55065	79.17044

ds - Forecast date

yhat - Predicted number of dengue cases

yhat_lower - Lower bound of the forecast interval

yhat_upperUpper - bound of the forecast interval

Example Interpretation

For 31 December 2026:

The Prophet model predicts approximately 75 dengue cases on 31 December 2026. Considering forecast uncertainty, the actual number of cases is expected to lie between 71 and 79 cases.

Overall Trend

From the forecast:

12 Dec 2026 → 64.65 cases

31 Dec 2026 → 75.35 cases

This indicates that:

Dengue cases are expected to increase gradually during the second half of December 2026.

The forecast rises from about 65 cases to 75 cases.

This suggests an upward trend in dengue incidence toward the end of the forecast horizon.

Forecast Uncertainty

Notice that: $yhat_lower < yhat < yhat_upper$

The interval is relatively narrow, indicating that the model is fairly confident about its predictions.

Report Interpretation

The Prophet model forecasted a gradual increase in dengue cases during December 2026. Predicted daily cases ranged from approximately 65 to 75 cases, with forecast intervals indicating moderate uncertainty. By 31 December 2026, the model predicted about 75 cases, with the actual number expected to fall between 71 and 79 cases. The results suggest a continuing upward trend in dengue incidence toward the end of the forecast period.